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A Blockchain-IoT Integrated Digital Vehicle Passport (DVP) System

راصد

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How it supports vision 2030?

VISION رؤية 2030 المملكة العربية السعودية KINGDOM OF SAUDI ARABIA

Rasid advances Saudi Vision 2030 "Digital Transformation" goals by establishing a tamper-proof, blockchain-integrated infrastructure for vehicular lifecycle management. By eliminating odometer fraud and information asymmetry through automated IoT validation, the system restores market trust and supports "Smart Mobility".

Introduction

Rasid is a Digital Vehicle Passport (DVP) system designed to combat fraud like odometer tampering and hidden accident records. By integrating IoT and Blockchain, it creates a secure, decentralized ledger that ensures absolute integrity throughout a vehicle's lifecycle. This transformation provides a transparent, trusted environment, ensuring every stage of a vehicle's history is documented and verifiable.

Objectives

- **System Integration:** Seamless integration of IoT and Blockchain for secure vehicle data recording.
- **Data Protection:** Advanced encryption and access control to protect sensitive information.
- **Lifecycle Management:** Tracking the complete digital lifecycle of vehicles from production to disposal.
- **Fraud Reduction:** Enhancing transparency to minimize fraud and financial risks.

Methodology

DATA LAYER

IoT Data Simulation — Vehicle Data Generation
- Mileage & fault codes
- OBD-II diagnostics

Python (IoT Simulator)

Preprocessing & Validation — Data Cleaning & Security
- Format validation
- AES Encryption
- SHA-256 Hashing

Python / Crypto Libraries

SYSTEM CORE

Backend Processing — API & Business Logic
- Request handling
- JWT Authentication
- RBAC Access Control

Node.js + Prisma

Data Storage — Secure Data Management
- Vehicle records storage
- Ledger metadata

PostgreSQL

Blockchain Simulation — Data Integrity Verification
- SHA-256 hashing
- Tamper detection

Hashing Algorithms

APPLICATION LAYER

Mobile Application — User Interface & Access
- View vehicle history
- Manage records

Flutter + Figma

User Output — Digital Vehicle Passport
- Verified vehicle data
- Transparent history

Final Output

Experiments

ARCHITECTURE COMPARISON

PUBLIC BLOCKCHAIN

Latency	~15 seconds
Cost	High (Gas Fees)
Security	Tamper-proof
IOT Integration	Complex

VS

HYBRID LEDGER (RASID)

Latency	< 1 second
Cost	Zero
Security	SHA-256 tamper-proof
IoT Integration	Seamless & scalable

Proposed solution:

The Hybrid Ledger architecture was adopted, which helped us in:
- Achieving real-time IoT data processing
- Zero transaction costs
- maintaining strict tamper-evident security.

Tools

Flutter
Prisma
TimescaleDB
JSON
Python
PostgreSQL
Node.js
Expres.js
SHA-265

Prototype

Conclusion

Rasid secures the automotive market by merging IoT and Blockchain into a "Digital Vehicle Passport". The system eliminates fraud and data fragmentation, ensuring total transparency. By providing a trusted, immutable record, it protects economic value and directly supports the digital goals of Saudi Vision.

Future Work

- **Government Integration:** linking with national traffic systems (Absher, Mojaz) for real-time vehicle data.
- **IoT & Blockchain Infrastructure**: deploying real IoT devices connected to a secure blockchain-based system.
- **AI Predictive Maintenance:** using AI to predict failures and reduce maintenance costs.

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